

TO SUPPLY LEFT-OVER FOOD TO POOR

Project Overview

FoodConnect is a Salesforce-powered platform designed to reduce food wastage while addressing hunger in underserved communities. The system connects surplus food providers such as restaurants, supermarkets, and event organizers with NGOs, shelters, and individuals in need. By leveraging Salesforce features like custom objects, workflows, and dashboards, the platform manages the entire process—from recording donations to assigning volunteers and ensuring on-time delivery. Donors can easily log their contributions, volunteers are guided with automated task allocation, and recipients receive food reliably, creating a transparent and efficient ecosystem.

The idea behind FoodConnect is to not only provide immediate relief from hunger but also establish a sustainable framework for long-term social good. By digitizing food redistribution, it ensures accountability, reduces environmental impact, and fosters a culture of sharing. With scalability at its core, the project can be replicated across different regions, making it adaptable to various communities. Ultimately, FoodConnect demonstrates how technology can be a force for positive change, bridging the gap between abundance and scarcity while promoting both social welfare and environmental sustainability.

2. Objectives

At the heart of FoodConnect is the mission to use technology to minimize food waste while improving food access for disadvantaged communities. The objectives are designed to deliver real-world value and establish a sustainable redistribution model.

Business Goals:

- A. **Optimize Food Redistribution:** Capture surplus food early and reallocate it efficiently to maximize its use.
- B. **Reduce Hunger Gaps:** Build a dependable chain from food donors to NGOs, shelters, and vulnerable groups.
- C. **Enhance Volunteer Productivity:** Provide structured roles, schedules, and digital tools to simplify participation.
- D. **Ensure Data Integrity and Trust:** Offer transparent tracking and reports to reinforce credibility with donors and recipients.

Specific Outcomes:

a. Integrated Platform:

- Centralize donor, venue, volunteer, and delivery data in one Salesforce system.
- Enable live tracking of donations from pickup to delivery.

- Provide stakeholders with real-time dashboards showing current operations.

b. Advanced Automation:

- Automatically assign volunteers based on food type, distance, and urgency.
- Trigger reminders and notifications to reduce delays in pickups and drop-offs.

c. Stronger Communication Channels:

- Enable direct updates between donors, volunteers, and recipients.
- Use automated alerts for schedule changes, confirmations, and emergencies.

d. Insight-Driven Decision Making:

- Analyze food movement patterns to identify bottlenecks and improve efficiency.
- Generate long-term metrics on waste reduced, meals provided, and volunteer contribution.

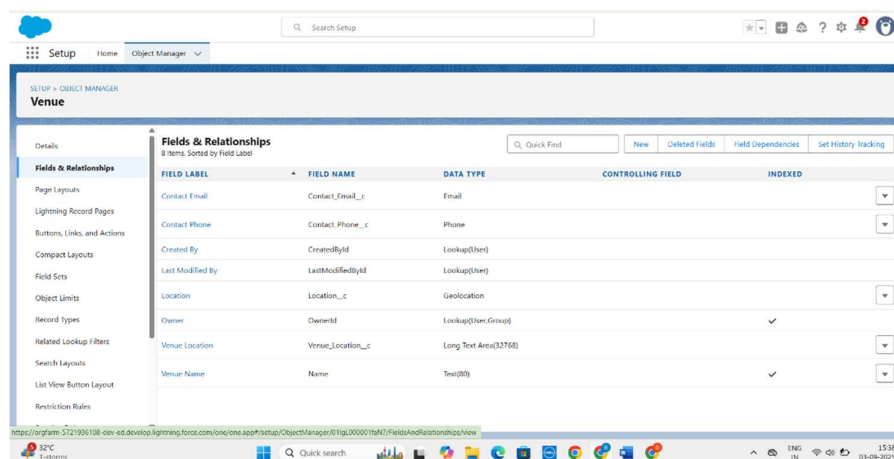
3. Salesforce Key Features and Concepts Utilized

FoodConnect makes use of Salesforce's ecosystem to design a streamlined and intelligent platform for food redistribution. Below are the major features and concepts applied:

Custom Objects

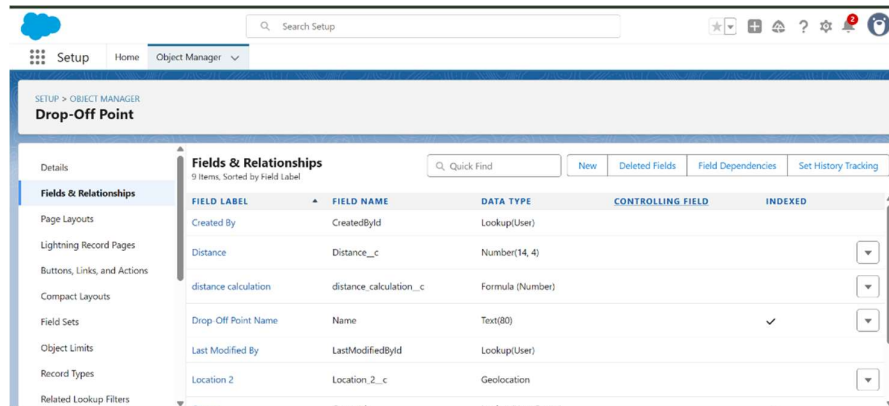
1. Venues:

- Stores details of restaurants, grocery stores, or event organizers providing surplus food.
- Includes attributes like location, capacity, type of food usually donated, and contact details.
- Helps create a structured donor database for efficient coordination.



2. Drop-Off Points:

- Represents shelters, NGOs, or distribution hubs where food is delivered.
- Tracks operational timings, storage capacity, and recipient details.
- Supports better planning of delivery routes and load distribution.

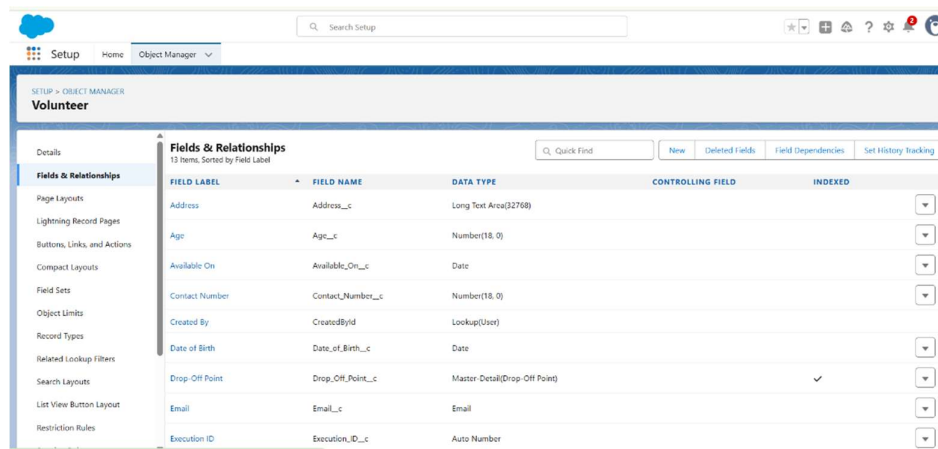


The screenshot shows the Salesforce Setup interface for the 'Drop-Off Point' object. The 'Fields & Relationships' tab is selected, displaying a table of 9 fields. The table has columns for Field Label, Field Name, Data Type, Controlling Field, and Indexed. The fields listed are: Created By (Lookup(User)), Distance (Number(14, 4)), distance calculation (Formula (Number)), Drop-Off Point Name (Text(80)), Last Modified By (Lookup(User)), and Location 2 (Geolocation). The 'Drop-Off Point Name' field is marked as indexed.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Distance	Distance_c	Number(14, 4)		
distance calculation	distance_calculation__c	Formula (Number)		
Drop-Off Point Name	Name	Text(80)		✓
Last Modified By	LastModifiedById	Lookup(User)		
Location 2	Location_2__c	Geolocation		

3. Volunteers:

- Maintains individual records with contact details, skill sets, and availability.
- Allows assignment of specific roles such as pickup, transport, or distribution.
- Builds a central registry for engaging and managing volunteer activities.



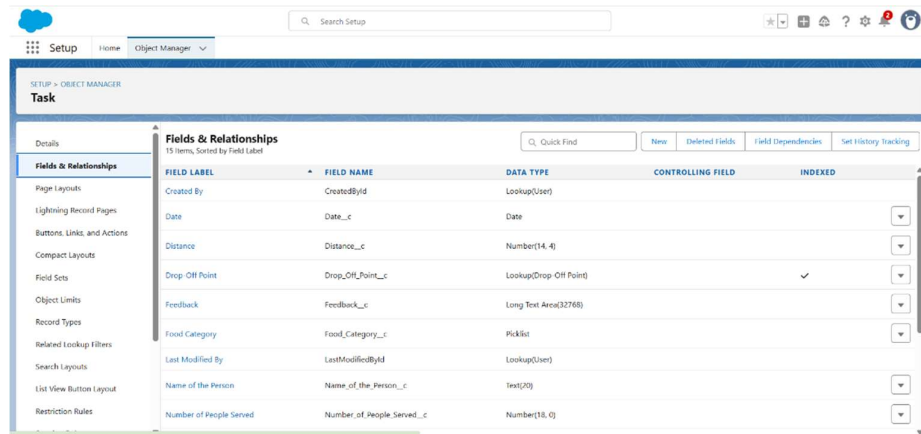
The screenshot shows the Salesforce Setup interface for the 'Volunteer' object. The 'Fields & Relationships' tab is selected, displaying a table of 15 fields. The table has columns for Field Label, Field Name, Data Type, Controlling Field, and Indexed. The fields listed are: Address (Long Text Area(32768)), Age (Number(18, 0)), Available On (Date), Contact Number (Number(18, 0)), Created By (Lookup(User)), Date of Birth (Date), Drop-Off Point (Master-Detail(Drop-Off Point)), Email (Email), and Execution ID (Auto Number). The 'Drop-Off Point' field is marked as indexed.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Address	Address__c	Long Text Area(32768)		
Age	Age__c	Number(18, 0)		
Available On	Available_On__c	Date		
Contact Number	Contact_Number__c	Number(18, 0)		
Created By	CreatedById	Lookup(User)		
Date of Birth	Date_of_Birth__c	Date		
Drop-Off Point	Drop_Off_Point__c	Master-Detail(Drop-Off Point)		✓
Email	Email__c	Email		
Execution ID	Execution_ID__c	Auto Number		

4. Tasks:

- Used to assign and monitor responsibilities like pickup, delivery, and food packaging.

- Tracks deadlines, responsible person, and current status of each activity.
- Ensures accountability and timely execution of redistribution steps.

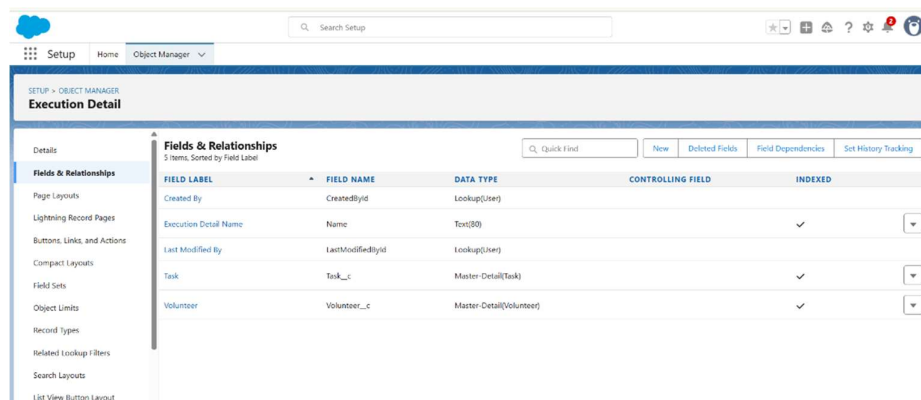


The screenshot shows the 'Fields & Relationships' section for the 'Task' object in Salesforce Setup. The left sidebar lists various setup options, and the main area displays a table of fields.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedBy	Lookup(User)		
Date	Date_c	Date		
Distance	Distance_c	Number(14, 4)		
Drop-Off Point	Drop_Off_Point_c	Lookup(Drop-Off Point)		✓
Feedback	Feedback_c	Long Text Area(32768)		
Food Category	Food_Category_c	Picklist		
Last Modified By	LastModifiedBy	Lookup(User)		
Name of the Person	Name_of_the_Person_c	Text(20)		
Number of People Served	Number_of_People_Served_c	Number(18, 0)		

5. Execution Detail:

- Logs completed actions, linking volunteers, venues, and drop-off points.
- Provides a history of deliveries and task outcomes for reporting purposes.
- Supports auditability and continuous improvement in operations.



The screenshot shows the 'Fields & Relationships' section for the 'Execution Detail' object in Salesforce Setup. The left sidebar lists various setup options, and the main area displays a table of fields.

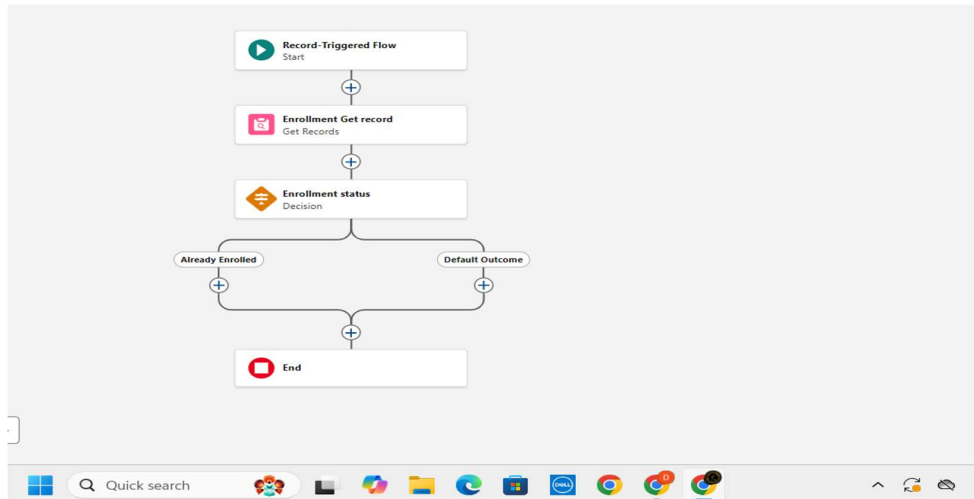
FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedBy	Lookup(User)		
Execution Detail Name	Name	Text(30)		✓
Last Modified By	LastModifiedBy	Lookup(User)		
Task	Task_c	Master-Detail(Task)		✓
Volunteer	Volunteer_c	Master-Detail(Volunteer)		✓

Relationships

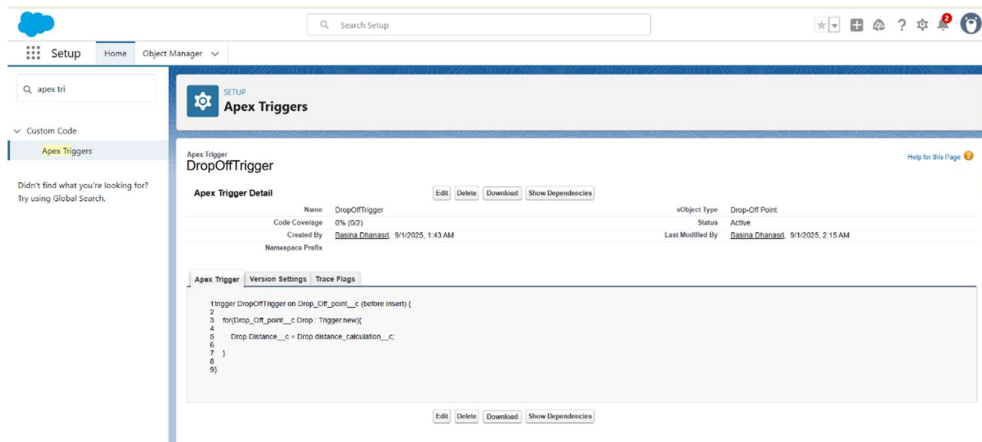
- **Master-Detail Relationships:** Ensure strong dependency, such as linking *Tasks* with *Execution Records* so that task closure always updates history.
- **Lookup Relationships:** Provide flexibility, for example connecting *Volunteers* with *Drop-Off Points* to allow assignments based on location and availability.

Automations and Logic

- **Flows:** Used to automate data entry and guide users when adding new donor venues or recording food pickups.



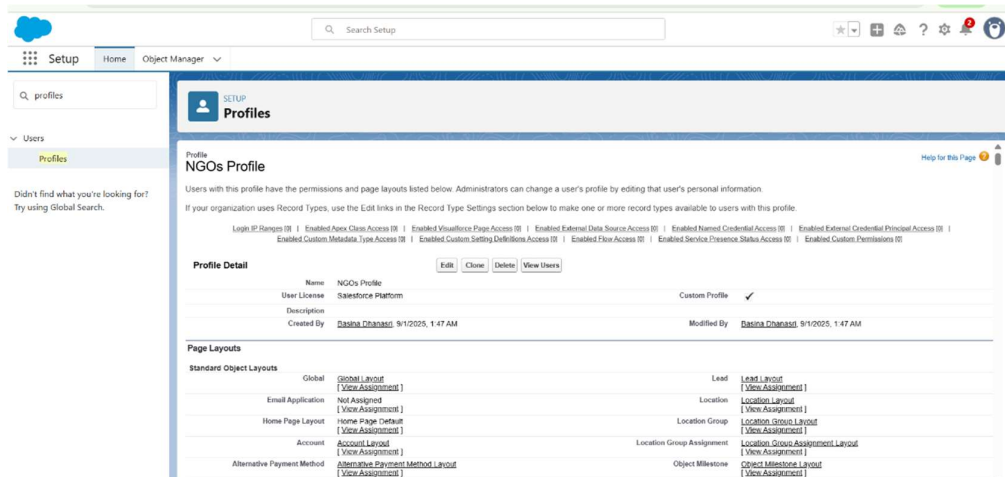
- **Apex Triggers:** Custom logic that calculates nearest volunteer to a venue and auto-assigns tasks, reducing delivery time and manual errors.



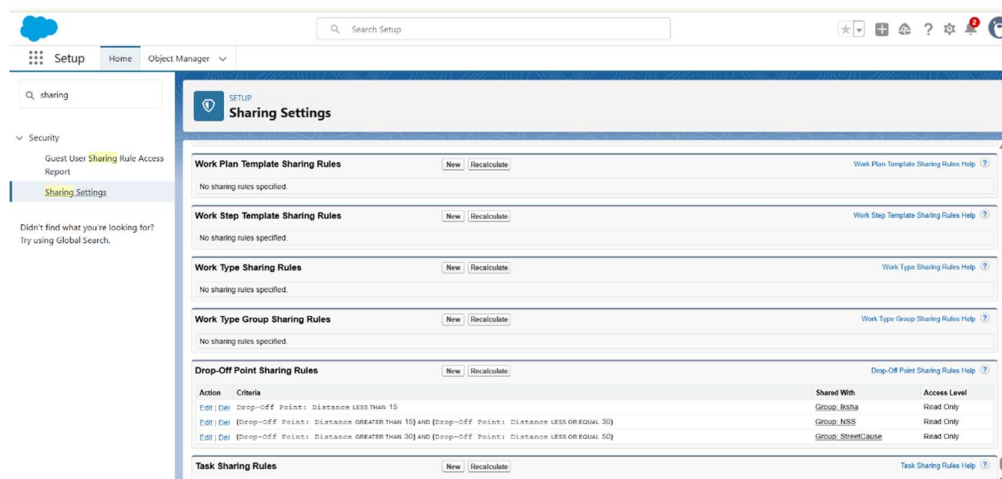
- **Workflow Rules & Notifications:** Sends automated alerts to volunteers about upcoming pickups, delays, or schedule changes.

Data Security and Sharing

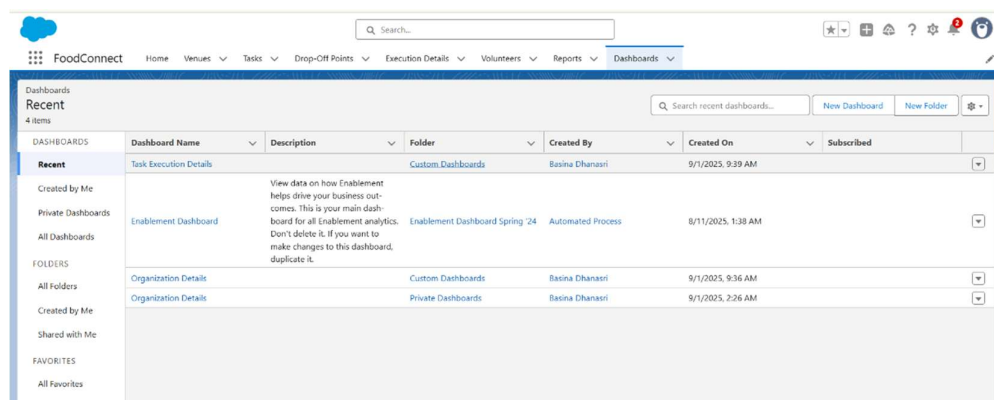
- **Profiles and Roles:** Different access levels for administrators, donors, and volunteers, ensuring sensitive donor/recipient data is protected.



- **Sharing Rules:** Allows controlled access to records so volunteers only view relevant assignments.



- **Dashboards:** Role-based dashboards give tailored insights—for example, donors can see food delivered, while admins track efficiency and impact.



User Experience Enhancements

- **Lightning App:** Custom-built FoodConnect app with simplified navigation across objects, reports, and dashboards.
- **Mobile-First Design:** Ensures volunteers can log updates directly from the field using smartphones.
- **Real-Time Notifications:** Keeps stakeholders informed about task status, food collection, and delivery updates instantly.

4. Detailed Steps to Solution Design

The FoodConnect platform was developed through a systematic approach to ensure it meets the operational needs of donors, volunteers, and recipients, while fully utilizing Salesforce's features. Below is the breakdown of the implementation:

Step 1: Developer Account Setup

- Created a Salesforce Developer Org to establish a dedicated workspace for building and testing the project.
- Activated all required features to access the complete Salesforce toolkit.

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First name: Basina ✓ Last name: Dhanasri ✓

Job title: Student ✓ Work email: 22kd1a0707@ler ✓

Company: Lendi ✓ Country/Region: India ✓

State/Province: Andhra Pradesh ✓

Your org may be provisioned on or migrated to Hyperforce, Salesforce's public cloud infrastructure.

☒ I agree to the Main Services Agreement - Developer Services and Salesforce Program Agreement. I acknowledge, as described in the Developer Documentation: (1) the Developer Edition includes autonomous and other generative AI features; and (2) Salesforce may limit use of those features and the org.

Step 2: Custom Object Creation

- Designed custom objects: Venue, Drop-Off Point, Volunteer, Task, and Execution Details.
- Configured appropriate data types and relationships to reflect real-world food redistribution workflows.

The screenshot shows the Salesforce Object Manager interface. The top navigation bar includes the Setup menu, Home link, and Object Manager dropdown. The main header displays the Object Manager title, a search bar, and buttons for Schema Builder and Create. Below the header, a table lists standard objects with columns for Label, API Name, Type, Description, and Last Modified. The objects listed are Account, Activity, Address, Agent Work, Alternative Payment Method, API Anomaly Event Store, Appointment Category, and Appointment Invitation.

LABEL	API NAME	TYPE	DESCRIPTION	LAS
Account	Account	Standard Object		
Activity	Activity	Standard Object		
Address	Address	Standard Object		
Agent Work	AgentWork	Standard Object		
Alternative Payment Method	AlternativePaymentMethod	Standard Object		
API Anomaly Event Store	ApiAnomalyEventStore	Standard Object		
Appointment Category	AppointmentCategory	Standard Object		
Appointment Invitation	AppointmentInvitation	Standard Object		

The screenshot shows the 'New Custom Object' configuration page in Salesforce. The page title is 'New Custom Object'. Below the title, there is a message about permissions. The main section is 'Custom Object Definition Edit', which includes fields for 'Custom Object Information'. The fields are: Label (with an example 'Account'), Plural Label (with an example 'Accounts'), Starts with vowel sound (checkbox), Object Name (with an example 'Account'), and Description.

Custom Object Information

The singular and plural labels are used in tabs, page layouts, and reports.

Label: Example: Account

Plural Label: Example: Accounts

Starts with vowel sound: ☐

The Object Name is used when referencing the object via the API.

Object Name: Example: Account

Description:

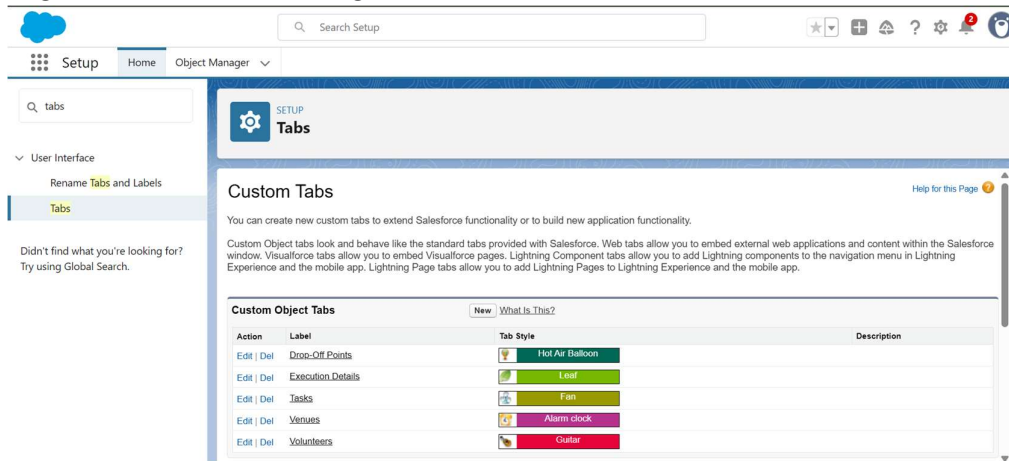
The screenshot shows the Salesforce Object Manager interface with a list of custom objects. The table has columns for Label, API Name, Type, Description, Last Modified, and Deployed. The objects listed are Execution Detail, Volunteer, Task, Drop-Off Point, and Venue.

LABEL	API NAME	TYPE	DESCRIPTION	LAST MODIFIED	DEPLOYED
Execution Detail	Execution_Detail__c	Custom Object		8/23/2025	✓
Volunteer	Volunteer__c	Custom Object		8/23/2025	✓
Task	Task__c	Custom Object		8/23/2025	✓
Drop-Off Point	Drop_Off_Point__c	Custom Object		8/23/2025	✓
Venue	Venue__c	Custom Object		8/23/2025	✓

Step 3: Custom Tabs for Navigation

- Added custom tabs for each object to improve accessibility.

- Organized tabs to make navigation seamless for end-users.



Step 4: Lightning App Development

- Built a Lightning App named *FoodConnect* using the App Manager.
- Integrated objects, reports, and dashboards into the app for centralized use.
- Assigned profiles to manage access based on user roles.

Step 5: Establishing Data Relationships

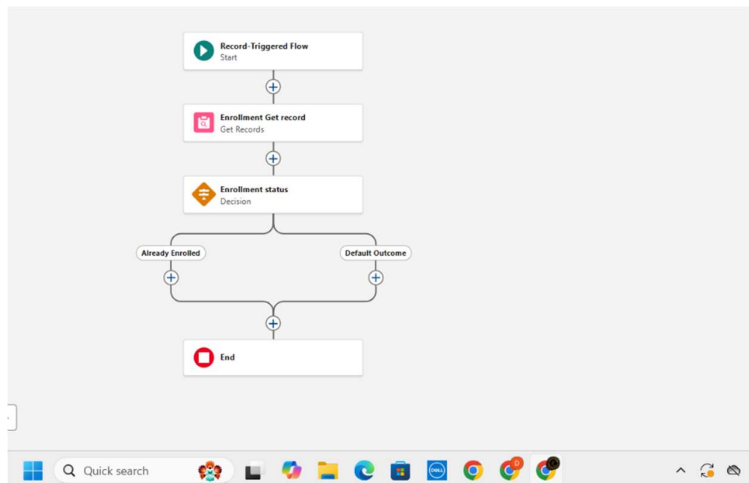
- Configured **Master-Detail** relationships (e.g., Tasks ↔ Execution Details) to enforce dependencies.
- Used **Lookup** relationships (e.g., Volunteers ↔ Drop-Off Points) for flexible task allocation.

Step 6: Field Configuration

- Added fields to capture donor, volunteer, and distribution details (e.g., Venue capacity, Task deadline, Volunteer availability).
- Applied validations to ensure clean and accurate data entry.

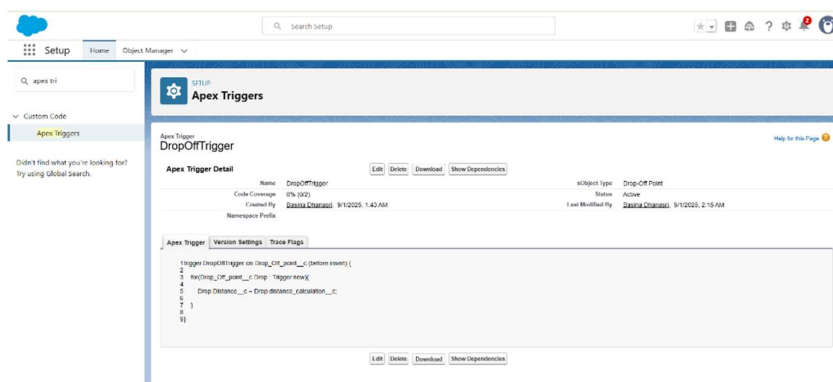
Step 7: Flow Design

- Created screen flows to guide users through adding donor venues and food records.
- Automated record creation processes to reduce errors and save time.



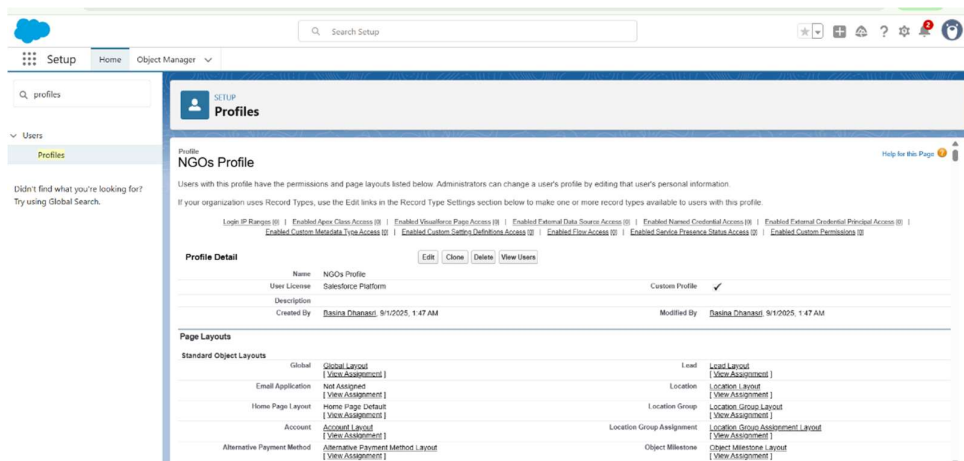
Step 8: Trigger Implementation

- Developed Apex triggers to calculate distances between donors and volunteers.
- Automated assignment of volunteers to tasks based on location and availability.



Step 9: User Management and Permissions

- Set up profiles for admins, donors, and volunteers with role-based access.
- Configured sharing rules to balance security with collaboration.



Step 10: Reporting and Dashboards

- Designed custom report types to analyze food distribution metrics and volunteer efficiency.
- Built dashboards to visualize key indicators like meals delivered, tasks completed, and waste reduction.

FoodConnect

Search...

Home Venues Tasks Drop-Off Points Execution Details Volunteers Reports Dashboards

Report: Venue with DropOff with Volunteer
venue and Drop Off point

Enable Field Editing

Total Records: 17 Total Distance: 7,050.8500

Volunteer Name	Venue Name	Drop-Off Point Name	Distance
- (10)	Community Function hall	Old age home	3,358.9400
	Green field Auditorium	Mother's old age Home	125.4000
	City Central	serve Food	103.2600
	City Central	DistributeFood	127.9200
	Sapphire Court	Helping Hands Shelter	128.4200
	a00gL00000IMUBZ	-	-
	a00gL00000IMXcT	-	-
	a00gL00000ISQdf	-	-
	Community Hall 2	-	-
	a00gL00000ISEHW	-	-

Row Counts Detail Rows Subtotals Grand Total

FoodConnect

Search...

Home Venues Tasks Drop-Off Points Execution Details Volunteers Reports Dashboards

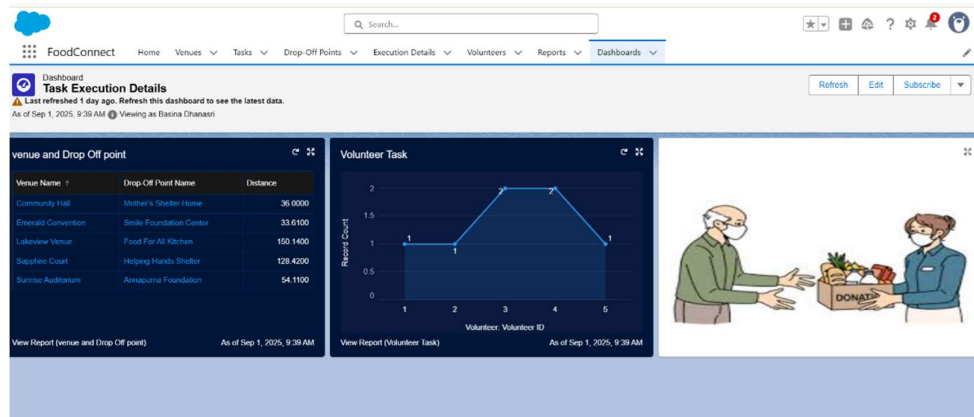
Report: Tasks with Execution Details and Volunteers
Volunteer Task

Enable Field Editing

Total Records: 10

Volunteer: Volunteer ID	Task: Task Name	Execution Detail: Execution Detail Name	Volunteer: Volunteer Name	Task: Owner Name	Rating	Date
1 (1)	Distribute Surplus Dinner - Hotel GreenPark	Execution4	Deepika	Basina Dhanasri	5	8/31/2025
Subtotal						
2 (1)	Dinner pickup	Execution5	Sahith	Basina Dhanasri	4	8/11/2025
Subtotal						
3 (2)	Dinner pickup	Execution3	Ramesh	Basina Dhanasri	4	8/11/2025
	Night Pickup	Execution7	Ramesh	Basina Dhanasri	4	8/4/2025
Subtotal						
4 (2)	Brunch Pickup	Execution2	amritha	Basina Dhanasri	3	7/23/2025
	Brunch Pickup	Execution6	amritha	Basina Dhanasri	3	7/23/2025
Subtotal						

Row Counts Detail Rows Subtotals Grand Total



5. Testing and Validation

To ensure FoodConnect operated smoothly and met all requirements, a structured testing approach was carried out. The process focused on both technical reliability and usability for stakeholders.

1. Unit Testing

- **Objective:** Confirm the accuracy and performance of individual components such as Apex triggers, flows, and custom objects.
- **Actions:**
 - Validated data entry forms for venues, tasks, and drop-off points to confirm accuracy and error handling.
 - Checked volunteer data storage and retrieval to ensure consistency and reliability.
 - Verified the correct execution of automated logic, such as volunteer assignment based on location.

2. User Interface Testing

- **Objective:** Assess the system's usability, design responsiveness, and accessibility for donors, volunteers, and administrators.
- **Actions:**
 - Tested page navigation and layout responsiveness to guarantee a seamless user experience.
 - Reviewed dashboards and reports to confirm that displayed data was correct, real-time, and easy to interpret.
 - Ensured cross-device compatibility, giving special focus to mobile access for on-field volunteers.

New Venue

Information

* Venue Name

HomeBell

* Contact Email

homebellgmail.com

Enter a valid email address, such as name@email.com.

Contact Phone

Location

Latitude

Longitude

Owner

Basina Dhanasri

6. Key Scenarios Addressed by Salesforce in the Implementation Project

The FoodConnect initiative uses Salesforce to solve practical challenges in food redistribution and volunteer coordination. Below are the key scenarios where Salesforce features provide effective solutions:

1. Recording Food Donation Venues

- **Scenario:** When a donor contributes surplus food, the system must capture venue details quickly and accurately.
- **Salesforce Solution:** Custom objects store venue data such as address, donor type, food capacity, and contact details, making it easy for administrators to manage and retrieve information.

2. Assigning and Monitoring Volunteer Tasks

- **Scenario:** Volunteers must be allocated duties like pickups and deliveries based on their schedules and proximity.
- **Salesforce Solution:** Relationships and automation assign tasks automatically, while the Task object monitors completion status and ensures accountability.

3. Managing Drop-Off Locations

- **Scenario:** Food must be transported efficiently to drop-off hubs where recipients can collect it.
- **Salesforce Solution:** Drop-Off Points are configured with fields for capacity, timings, and storage conditions, allowing smooth coordination and balanced distribution.

4. Improving Volunteer Participation

- **Scenario:** Volunteers need visibility into their work, including upcoming assignments and completed contributions.

- **Salesforce Solution:** The Volunteer object provides a self-service interface where members can update availability, review tasks, and track their impact.

5. Ensuring Real-Time Communication

- **Scenario:** Donors, volunteers, and admins require timely updates on changes, delays, or new tasks.
- **Salesforce Solution:** Automated alerts and workflow notifications deliver instant updates, ensuring smooth communication and reducing confusion.

6. Monitoring Impact Through Reports

- **Scenario:** Stakeholders want to measure food saved, meals delivered, and volunteer participation.
- **Salesforce Solution:** Reports and dashboards generate live analytics, providing insights into redistribution efficiency and highlighting areas for improvement.

7. Automating Logistics Calculations

- **Scenario:** Delivery efficiency depends on assigning the nearest volunteer to a donation site.
- **Salesforce Solution:** Apex logic calculates distances and optimizes task allocation, reducing travel time and improving resource utilization.

8. Enforcing Secure Access

- **Scenario:** Each role—admin, donor, or volunteer—requires access to only the data relevant to them.
- **Salesforce Solution:** Profiles, roles, and sharing settings restrict data visibility while ensuring collaboration where needed.

Conclusion

FoodConnect demonstrates how Salesforce can be leveraged to fight food waste and hunger through technology-driven solutions. The platform streamlines food collection and delivery, empowers volunteers with structured workflows, and ensures transparency through real-time dashboards. Automation reduces manual work, while reports provide insights into community impact. With its scalable design, FoodConnect serves as a replicable model for sustainable food redistribution in diverse regions.

